Worksheet — Angled Launch Simulation

Use with *Simulation - Angled launch.html* · A.1 Kinematics: projectile launched at an angle — components of *u*, maximum height, time of flight, range; goes with the launcher / water-jet range demo

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/12-angled-launch** in a browser (offline: *Simulation - Angled launch.html*). Set the speed *u*, the angle θ and the launch height *h*, then ▶ **Launch**. With **Keep paths** on, earlier launches stay on the screen so you can compare angles.

Part 1 before the lab (≈ 15 min) · **Part 2** during the demo (≈ 15 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81$ m s^{-2}. Resolve first: $u_x = u \cos\theta$, $u_y = u \sin\theta$. Then horizontal $a_x = 0$, vertical $a_y = -g$.

Scan to open

Part 1 — Pre-lab: predict, then test

Write your prediction **before** you run the simulation. Then test it and explain what you saw.

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Same speed, from the ground: which angle gives the longest range?			
2. Compare 30° and 60° (same speed). How do the ranges compare?			
3. What is the velocity of the ball at the highest point?			
4. Which angle (15°, 45° or 75°) keeps the ball in the air longest?			

5. Calculate the time of flight for u = 10 m s^{-1} at 30° from the ground. Check with the simulation.

Part 2 — During the lab: the range demo

Demo. The launcher (or water jet) fires at the same speed at different angles; mark where it lands.
Simulation: u = 6.0 m s^{-1} from the ground, same angles, Keep paths on, ● **Record** each launch.

θ / °	*R* (demo) / m	*R* (simulation) / m	difference / %
15			
30			
45			
60			
75			

1. Range at 45° in the simulation (u = 6.0 m s^{-1}). Show the calculation.

2. Which pairs of angles land at the same place? Explain using $R = u^2 \sin 2\theta / g$.

3. In the demo the longest range was 3.0 m, at about 45°. Estimate the launch speed of the demo.

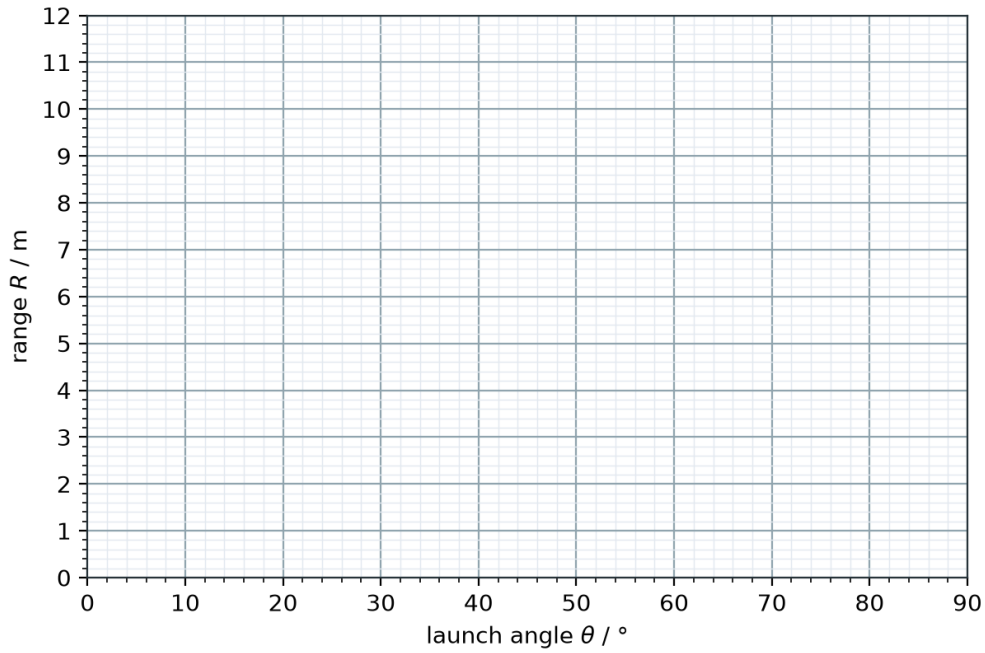
4. Why are the demo ranges a little shorter than the simulation, especially at steep angles? Give two reasons.

Part 3 — Homework: analyse

u = 10 m s^{-1} from the ground. Record one launch at each angle.

θ / °	10	20	30	40	50	60	70	80
R / m								

1. Plot R (y-axis) against θ (x-axis) and draw a smooth curve.



2. Maximum range from your graph: R = _____ m at θ = _____°

3. A football is kicked from the ground at 15 m s^{-1}, 35° above the horizontal. Calculate the maximum height.

4. Calculate the time of flight.

5. Calculate the range.

6. What is the speed of the ball at the highest point?

7. A stone is thrown from the top of a 20 m cliff at 12 m s^{-1}, 30° above the horizontal. How far from the foot of the cliff does it land?

8. From the top of the cliff, is the best angle for the longest range 45°, more or less? Explain.